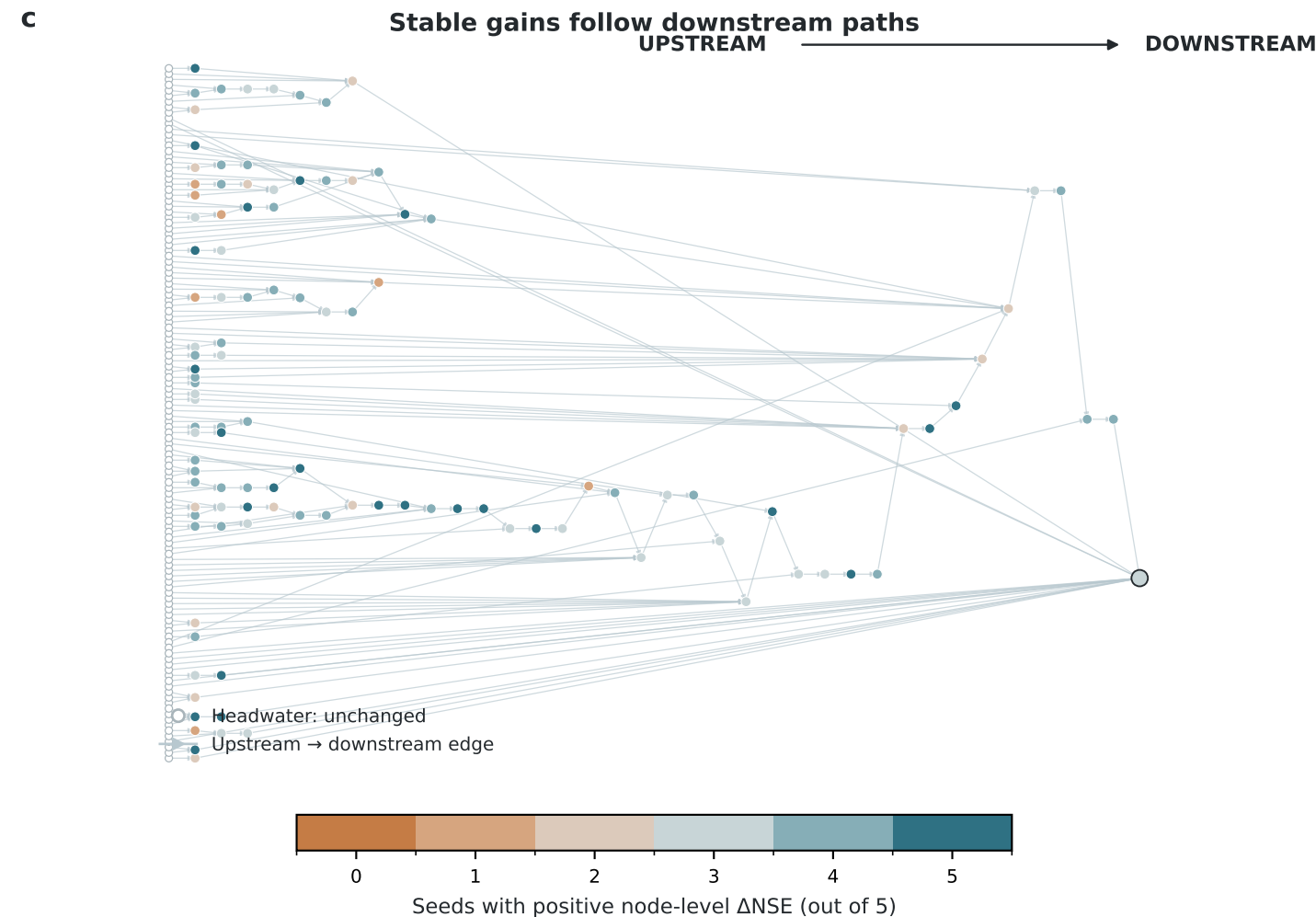
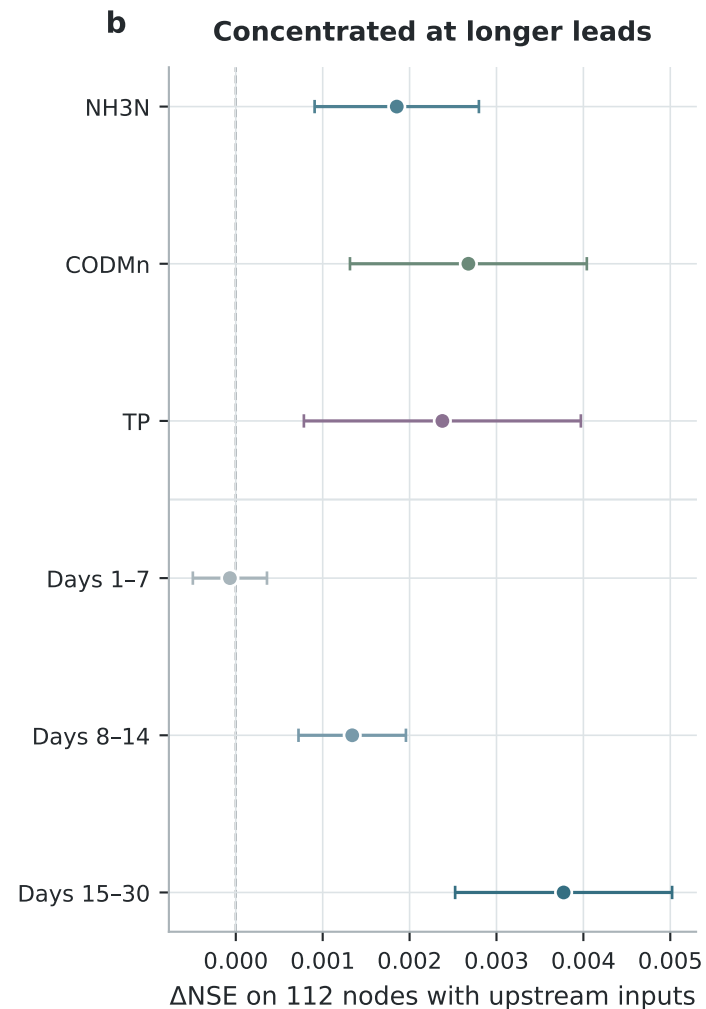
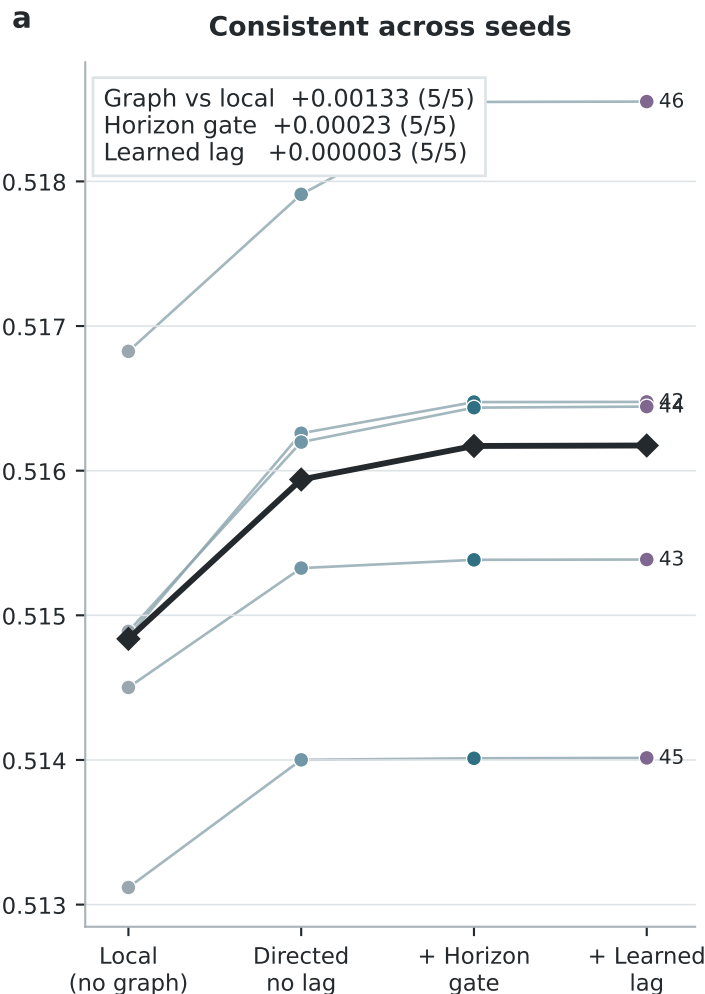


Directed upstream information improves validation; explicit lag adds little

Five paired seeds · 238 monitored segments · 237 directed edges · all refinement stages strictly nest their frozen predecessor



Validation split only; the held-out test set remains unopened. Node colors show predictive increments, not causal effects or recovered travel times.